WENYIN (KNUT) GU

ELECTRICAL ENGINEERING / DIGITAL SIGNAL PROCESSING / MACHINE LEARNING

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01 EDUCATION

University of California, Irvine · *B.S. in Electrical Engineering, Specialization: DSP* 2025 – PRESENT
GPA **3.64 / 4.0** ▪ Dean's List (all quarters) ▪ **IEEE-Eta Kappa Nu** 2026

Cañada College · *A.S. in Engineering* 2022 – 2025
GPA **4.0 / 4.0** ▪ **Valedictorian** ▪ ***Summa Cum Laude*** ▪ **Phi Theta Kappa** 2023 ▪ Dean's List (all semesters)

Stanford University · *Fundamentals of Data Science (4-week intensive)* SUMMER 2023
TOPICS Python, statistics, mathematical optimization, linear algebra, responsible AI

02 RESEARCH

Recovering and Combining Partial Multi-Camera Evidence under Wi-Fi Jamming JUL 2026
EECS 199 Independent Study · Advised by Prof. Syed A. Jafar

- Modeled Wi-Fi jamming as uplink packet erasure and built a 5-camera, 48-step Python simulation (galois, NumPy, OpenCV) splitting each record into coarse and fine Lagrange-coded layers over GF(2003), with T-privacy masking.
- Layered coding recovered 94% of usable evidence versus 36% uncoded at 3.9× transmission overhead; cross-camera fusion outperformed the best single camera on both accuracy and coverage.

"Agentic Affordance: Why LLMs Invite the Intentional Stance" MAY 2026
With Aileen C. Luo · *6th International Conference on Philosophy of Mind: Artificial Intelligence (6ICPH)*

- Developed the notion of "agentic affordance" to explain why human–LLM interaction warrants an intentional stance despite a lack of mechanistic interpretability.

03 PROJECTS

Diffusion-Based CSI Recovery for OFDM Systems · *Senior Design, Team Lead* IN PROGRESS

- Building a Python OFDM simulation comparing interpolation, LMMSE, and a conditional diffusion model (DDPM) for reconstructing CSI from sparse, noisy pilots under random, periodic, and burst loss, evaluated by NMSE and BER.
- Developing an acoustic OFDM prototype with self-built TX/RX and jammer circuits for an Arduino-controlled live demo.

Spatial vs. Temporal Edge Detection Simulation MAR 2026

- Built Python simulations (NumPy/SciPy) comparing 2D spatial convolution (Sobel operators) with temporal high-pass filtering (difference equations) for edge detection.
- Showed that event-based temporal filtering achieves superior signal sparsity and efficiency in motion tracking by mathematically suppressing static backgrounds to zero.

Music-Reactive LED System MAR 2026

- Designed a custom microphone preamplifier and 3-band active analog crossover (LM358 op-amps) to separate bass, midrange, and treble in hardware.
- Programmed a non-blocking Arduino state machine and interactive OLED GUI that turn multi-channel ADC input into three real-time LED visualization modes.

04 EXPERIENCE & LEADERSHIP

Communications Subsystem Engineer · *Legacy Robotics, UC Irvine* APR 2026 – PRESENT

- Set up and maintain the rover's 5 GHz video link (Ubiquiti airMAX) and 900 MHz telemetry link (RFD radios, Yagi base station, Mission Planner) for the University Rover Challenge, covering antenna selection, pairing, and field testing under competition rules on fast setup, FCC compliance, sub-band switching, and non-line-of-sight range.

Webmaster · *IEEE-Eta Kappa Nu (HKN), UC Irvine* MAY 2026 – PRESENT

- Maintain and architect the chapter website, managing updates to event schedules, membership rosters, and technical archives.

Mathematics Tutor · *Cañada College Learning Center* 2022 – 2024

- Provided one-on-one and drop-in tutoring, guiding students through problem-solving and analytical reasoning.

Community Service · *Shanghai, China* 2007, 2010

- Water Supply Team Lead, Shanghai World Expo (2010): led safety inspections of water-supply networks safeguarding drinking water for 73 million visitors over six months.
- Volunteer, Special Olympics World Summer Games (2007).

05 SKILLS

PROGRAMMING	Python, MATLAB, C++, Verilog
HARDWARE & EDA	Analog circuit design (op-amps, active filters, BJT/MOSFET), Cadence Virtuoso, SolidWorks, Onshape, Arduino/ESP32 integration (ADC, I2C, OLED)
COMMUNICATIONS	RF link budgeting, antenna selection, OFDM channel estimation, erasure/Lagrange coded computing